BRIGHTLEARN

BIGQUERY PRACTICAL

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QUESTION 1:

```sql
SELECT *
FROM`retail-479102`.Sales.Data
WHERE
EXTRACT(YEAR FROM Date) = 2023;
```

This query will process 72.69 KB when run.

Using on-demand processing quota

Query results

Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	Transaction ID	Date	Customer ID	Gender	Age
1	191	2023-10-18	CUST191	Male	

QUESTION 2:

```sql
-- Expected output: All columns
SELECT *
FROM`retail-479102`.Sales.Data
WHERE `Total Amount` > (
   SELECT AVG(`Total Amount`)
   FROM`retail-479102`.Sales.Data
);
```

Query completed

Using on-demand processing quota

Query results

Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category
1	21	2023-01-14	CUST021	Female	50	Beauty
2	28	2023-04-23	CUST028	Female	43	Beauty
3	128	2023-07-05	CUST128	Male	25	Beauty

QUESTION 3:

```
-- Q3. Calculate the total revenue (sum of Total Amount).
-- Expected output: Total_Revenue
SELECT
    SUM(`Total Amount`) AS Total_Revenue
FROM`retail-479102`.Sales.Data;
```

This script will process 153.19 KB when run.

Query results

Job information | Results | Visualisation | JSON | Exec

Row	Total_Revenue
1	456000

QUESTION 4:

```
-- Q4. Display all distinct Product Categories in the dataset.
-- Expected output: Product_Category
SELECT DISTINCT(`Product Category`) AS Product_Category
FROM`retail-479102`.Sales.Data;
```

Query completed

Using on-demand processing quota

Query results

Job information | Results | Visualisation | JSON | Execution details

Row	Product_Category
1	Beauty
2	Clothing
3	Electronics

QUESTION 5:

```sql
-- Expected output: Product_Category, Total_Quantity
SELECT `Product Category`,
       SUM(Quantity) AS Total_Quantity
FROM`retail-479102`.Sales.Data
GROUP BY `Product Category`;


```

This script will process 181.34 KB when run.

Using on-demand processing quota

Query results

Job information | Results | Visualisation | JSON | Exec

Row	Product Category	Total_Quantity
1	Beauty	771
2	Clothing	894
3	Electronics	849

QUESTION 6:

```sql
-- Expected output: Customer_ID, Age, Age_Group
SELECT `Customer ID`,
      Age,
CASE
      WHEN Age < 30 THEN 'Youth'
      WHEN Age BETWEEN 30 AND 59 THEN 'Adult'
      WHEN Age > 59 THEN 'Senior'
      END AS Age_Group
```

Query completed

Using on-demand processing quota

Query results

Job information | Results | Visualisation | JSON | Execution details | Exe

Row	Customer ID	Age	Age_Group
1	CUST191	64	Senior
2	CUST204	39	Adult
3	CUST230	54	Adult

QUESTION 7:

```
-- Expected output: Gender, High_Value_Transactions
SELECT Gender,
      COUNT(`Total Amount`) AS High_Value_Transactions
FROM`retail-479102`.Sales.Data
WHERE `Total Amount` > 500
GROUP BY Gender;
```

Query completed

Query results

Job information | Results | Visualisation | JSON | Execut

Row	Gender	High_Value_Tran...
1	Female	155
2	Male	144

QUESTION 8:

```
      SUM(`Total Amount`) AS Total_Revenue
FROM`retail-479102`.Sales.Data
GROUP BY Product_Category
Having Total_Revenue > 5000;
```

This script will process 230.59 KB when run.

Using on-demand processing quota

Query results

Job information | Results | Visualisation | JSON

Row	Product_Category	Total_Revenue
1	Beauty	143515
2	Clothing	155580
3	Electronics	156905

QUESTION 9:

```
SELECT `Transaction ID`,
       `Price per Unit`,
CASE
      WHEN `Price per Unit` < 50 THEN 'Cheap'
      WHEN `Price per Unit` BETWEEN 50 AND 200 THEN 'Moderate'
      WHEN `Price per Unit` > 200 THEN 'Expensive'
      END AS Unit_Cost_Category
```

Query completed

Using on-demand processing quota

Query results

Job information | Results | Visualisation | JSON | Execution details

Row	Transaction ID	Price per Unit	Unit_Cost_Category
1	191	25	Cheap
2	204	25	Cheap
3	230	25	Cheap

QUESTION 10:

```
CASE
      WHEN `Total Amount` > 1000 THEN 'High'
      ELSE 'Low'
      END AS Spending_Level
FROM`retail-479102`.Sales.Data
WHERE Age > 40;
```

Query completed

Using on-demand processing quota

Query results

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	Customer ID	Age	Total Amount	Spending_Level
1	CUST191	64	25	Low
2	CUST230	54	25	Low
3	CUST232	43	25	Low